

1 DATA ACQUISITION AND PREPROCESSING STRATEGY

In the proposed multi-objective joint-topology and parameter estimation (*MO-JTPE*) strategy, the network information, comprising voltage magnitudes and power-injection measurements, is required as input. In this paper, a strategy is proposed for data acquisition and preprocessing using the Backward-Forward Sweep (BFS) distribution network power-flow simulation [1]. The resulting data can be classified into two categories: the input dataset and the target dataset. The input dataset, D^{In} , comprises the voltage magnitude, V , active power injection, P , reactive power injection, Q , and sample measurements at all network nodes. The target dataset, representing the expected output dataset, D^{Out} , comprises the voltage angle, θ , topology, $\mathbf{T}$, resistance, R , and reactance, X , for all nodes/branches in the network. The proposed data acquisition and pre-processing strategy is achieved using Algorithm S1.

Algorithm S1 Data Acquisition and Pre-processing (*DAP*)

Inputs: Network information: line data comprising the branch sending-end and receiving-end, node identification numbers (IDs); the corresponding resistance and reactance values; the bus data comprising receiving-end node identification numbers; the corresponding active and reactive power generations and loadings.

Outputs: Input dataset $\mathbf{D}^{\text{In}} = \{\mathbf{V}, \mathbf{P}, \mathbf{Q}\}$ for the *MOJTPE* and target dataset $\mathbf{D}^{\text{Out}} = \{\mathbf{\Theta}, R, X, \mathbf{T}\}$, reserved for performance analysis.

Steps:

1. Obtain network line and bus data and initialize the scenario counter $s = 0$.
2. Extract the topology matrix $\mathbf{T}_0$, resistance vector R , reactance vector X , load vectors P^{L0} , Q^{L0} , and generation vectors P^{G0} , Q^{G0} .
3. Standardize node numbering and convert quantities into per-unit values to obtain:

$$\begin{aligned}
P^{L0} &= [p_1^{L0}, \dots, p_{N^B}^{L0}]^T, \\
Q^{L0} &= [q_1^{L0}, \dots, q_{N^B}^{L0}]^T, \\
R &= [r_1, \dots, r_{N^B-1}]^T, \\
X &= [x_1, \dots, x_{N^B-1}]^T, \\
\mathbf{T} &= \begin{bmatrix} t_{1,1} & \cdots & t_{n-1,1} & \cdots & t_{N^B-1,1} \\ 2 & \cdots & n & \cdots & N^B \end{bmatrix}^T
\end{aligned}$$

4. Initialize a new scenario: $s = s + 1$.
5. For each of the N^B buses, integrate load uncertainty according to (1) and (2).
6. For each of the N^B buses, integrate generation uncertainty according to (3) and (4).
7. For each of the N^B buses, compute nominal active and reactive power injections using (5) and (6).
8. Execute the BFS power flow method (Algorithm S2) to evaluate nominal voltage magnitude V^0 and angle $\theta[1]$.
9. For each of the N^B buses, integrate measurement uncertainty according to (7), (8), and (9).
10. Update and store the resulting $\mathbf{V}$, $\mathbf{\Theta}$, $\mathbf{P}$, and $\mathbf{Q}$ datasets in the data bank.
11. If $s < N^S$, return to **step 4**; otherwise, proceed to **step 12**.
12. Construct the input dataset $\mathbf{D}^{\text{In}} = \{\mathbf{V}, \mathbf{P}, \mathbf{Q}\}$ and the target dataset $\mathbf{D}^{\text{Out}} = \{\mathbf{\Theta}, R, X, \mathbf{T}\}$.

Load uncertainty (Algorithm S1, step 5):

$$p_{n,s}^L = \left(\frac{1}{s} + \eta_c^L + \eta_v^L \mu_{n,s}^{PL} \right) s p_n^{L0}, \quad (1)$$

$$q_{n,s}^L = \left(\frac{1}{s} + \eta_c^L + \eta_v^L \mu_{n,s}^{QL} \right) s q_n^{L0}, \quad (2)$$

where $n = 1, 2, 3, \dots, N^B$ and $s = 1, 2, 3, \dots, N^S$ denote the node and scenario counters, respectively; η_c^L and η_v^L represent the percentages of constant and variable loads; and $-1 \leq \mu^{PL} \leq 1$ and $-1 \leq \mu^{QL} \leq 1$ are the active and reactive load uncertainty coefficients, respectively.

Generation uncertainty (Algorithm S1, step 6):

$$p_{n,s}^G = \left(\frac{1}{s} + \eta_c^G + \eta_v^G \mu_{n,s}^{PG} \right) s p_n^{G0}, \quad (3)$$

$$q_{n,s}^G = \left(\frac{1}{s} + \eta_c^G + \eta_v^G \mu_{n,s}^{QG} \right) s q_n^{G0}, \quad (4)$$

where η_c^G and η_v^G denote the percentages of constant and variable generation, respectively, and μ^{PG} and μ^{QG} represent the active and reactive power generation uncertainty coefficients. Equations (1)-(4) are used to simulate uncertainties in active and reactive loads and generation. The variable 's' is a scenario counter representing time instants that are not necessarily consecutive. When closely observed, it should be noted that the real and reactive power loads at bus 'n' are being added with a constant and a variable uncertainty factor that also depend on the time instants (the scenario counter), thus adding further uncertainty.

Computing nominal active and reactive power injections (Algorithm S1, step 7):

$$p_{n,s}^0 = p_{n,s}^L - p_{n,s}^G \quad (5)$$

$$q_{n,s}^0 = q_{n,s}^L - q_{n,s}^G \quad (6)$$

Measurement uncertainty (Algorithm S1, step 9):

$$p_{n,s} = \left(1 + \sigma_N^2 \mu_{n,s}^{NP} \right) p_{n,s}^0, \quad (7)$$

$$q_{n,s} = \left(1 + \sigma_N^2 \mu_{n,s}^{NQ} \right) q_{n,s}^0, \quad (8)$$

$$v_{n,s} = \left(1 + \sigma_N^2 \mu_{n,s}^{NV} \right) v_{n,s}^0, \quad (9)$$

where σ_N^2 is the generalized measured noise covariance, and $-1 \leq \mu_{n,s}^{NV} \leq 1$, $-1 \leq \mu_{n,s}^{NP} \leq 1$, and $-1 \leq \mu_{n,s}^{NQ} \leq 1$ are the measurement uncertainty coefficients for voltage, active, and reactive power injection measurements, respectively.

The resulting P and Q datasets are each in the form of an $N^B \times N^S$ matrix, where N^B and N^S denote the number of nodes and samples, respectively. Whereas, $\mathbf{R}$ and $\mathbf{X}$ are column vectors, each with $(N^B - 1)$ elements. $\mathbf{T}$ is a matrix with $(N^B - 1)$ rows and 2 columns. For a particular network, the topology matrix $\mathbf{T}$ is comprised of node numbering, with the first column consisting of a column vector of sending end nodes, and the second column having the corresponding receiving end nodes. The elements

in the second column of $\mathbf{T}$ are a set of consecutive integers that begin with 2 and are sorted in ascending order. As described earlier, each sample of the network states, V^0 and θ , is obtained using the BFS power-flow technique described in Algorithm S2.

Algorithm S2 Backward-Forward Sweep (BFS) Power Flow Technique

Inputs: Network parameter vectors (R, X) , topology matrix $\mathbf{T}$, sample active and reactive power vectors (P, Q)

Outputs: The network s^{th} sample reference states $V_s = [v_{2,s}, v_{3,s}, \dots, v_{k,s}, \dots, v_{N^B,s}]^T$ and voltage angle vector $\theta_s = [\theta_{2,s}, \theta_{3,s}, \dots, \theta_{k,s}, \dots, \theta_{N^B,s}]^T$.

Steps:

1. Obtain and prepare the network parameters (R, X) , topology matrix $\mathbf{T}$, and power injections (P, Q) .
 2. Construct $(N^B - 1) \times (N^S - 1)$ the branch connectivity matrix $\mathbf{A}_{k,m}^{-1}$, which is the inverse of the Bus Incidence Branch Connection (BIBC) matrix, by evaluating its elements using (10) [2]. *Note: Bus 1 is the slack bus and is eliminated. Therefore, only the load-bus states are considered in the power flow.*
 3. Initialize voltage tolerance $\varepsilon = 10^{-5}$; and set $v_{k,s} = 1$, and $\theta_{k,s} = 0$ for $k = 1, 2, 3, \dots, (N^B - 1)$.
 4. For each of the $(N^B - 1)$ receiving-end buses, estimate the injected current using (11). *Note: k is used for branch numbering, and as such, $k = n - 1$*
 5. Update the receiving-end complex voltage vector using (12) in addition to (13).
 6. Repeat **step 5**, to re-evaluate the final vector of branch voltage drop dv_s^2 .
 7. Update the voltage angles across all nodes using (14).
 8. Update the voltage magnitude across all nodes using (15).
 9. Determine the absolute error in the branch voltage drops $\epsilon = |dv_s^1 - dv_s^2|$.
 10. Find the bus b having the highest deviation and set $\epsilon = \epsilon_b$.
 11. Check if $\epsilon_{max} > \varepsilon_b$, if so, return to **step 5**. Otherwise, move to the next step (**step 12**).
 12. Output s^{th} voltage magnitude vector, $|V_s|$, and angle vector, θ_s , respectively.
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The inverse of the incidence branch connectivity matrix (BIBC) (Algorithm S2, step 2) given in (10)[1],

$$(\mathbf{A}^{-1})_{k,m} = \begin{cases} 1, & k = m \\ -1, & T_{k,1} = k + 1 \text{ and } T_{k,2} = m + 1 \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

Calculating bus injection current (Algorithm S2, step 4) given in (11)[1],

$$i_{k,s} = \frac{p_{k,s} - jq_{k,s}}{v_{k,s}(\cos\theta_{k,s} + j\sin\theta_{k,s})} \quad (11)$$

Evaluate the initial vector of branch voltage drop (Algorithm S2, step 5) given in (12) [1],

$$dv_s^1 = [\mathbf{A}^T][\mathbf{Z}][\mathbf{A}][I_s], \quad (12)$$

where $\mathbf{Z}$ is given by (13)[1],

$$\mathbf{Z} = \begin{bmatrix} r_1 + jx_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_2 + jx_2 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & r_k + jx_k & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & r_{NB-1} + jx_{NB-1} \end{bmatrix} \quad (13)$$

where r_k and x_k are the resistance and reactance of the network branches, respectively, and $I_s = [\dot{i}_{1,s} \ \dot{i}_{2,s} \ \cdots \ \dot{i}_{k,s} \ \cdots \ \dot{i}_{NB,s}]^T$

Update the voltage angles across all nodes (Algorithm S2, step 7):

$$\theta_{k,s} = \tan^{-1} \left(\frac{\text{imag}(1 - dv_{k,s}^2)}{\text{real}(1 - dv_{k,s}^2)} \right) \quad (14)$$

Update the voltage magnitude across all nodes (Algorithm S2, step 8):

$$v_{k,s} = \sqrt{\text{imag}(1 - dv_{k,s}^2)^2 + \text{real}(1 - dv_{k,s}^2)^2} \quad (15)$$

Table S1: Statistical Information of the IEEE 30-bus Network Input Datasets

Parameter	Indices	ISM	RF	RF+DG	NST+DG
V (p.u.)	Min	0.910	0.846	0.832	0.840
	Mean	0.950	0.938	0.938	0.947
	Max	1.000	1.000	1.000	1.010
	Std. Dev.	0.024	0.033	0.034	0.035
Load P (MW)	Mean	4.401	5.846	6.830	7.964
	Max	4.973	8.705	14.661	14.808
	Std. Dev.	0.439	1.406	2.132	2.301
Load Q (MVar)	Mean	10.698	11.422	11.347	11.380
	Max	12.931	15.191	19.919	20.118
	Std. Dev.	1.389	1.773	1.952	2.163
DG P (MW)	Mean	—	—	0.957	2.002
	Max	—	—	5.935	5.995
	Std. Dev.	—	—	1.454	1.546
DG Q (MVar)	Mean	—	—	0.194	0.406
	Max	—	—	2.684	2.711
	Std. Dev.	—	—	0.440	0.572

Table S2: Statistical Information of the IEEE 33-bus Network Input Datasets

Parameter	Indices	ISM	RF	RF+DG	NST+DG
V (p.u.)	Min	0.900	0.828	0.822	0.830
	Mean	0.944	0.931	0.931	0.940
	Max	1.000	1.000	1.000	1.010
	Std. Dev.	0.034	0.043	0.044	0.045
Load P (MW)	Mean	3.057	3.533	3.964	4.509
	Max	4.083	5.097	9.106	9.197
	Std. Dev.	0.455	0.633	1.077	1.274
Load Q (MVar)	Mean	3.423	4.335	4.304	4.310
	Max	3.838	5.847	5.847	5.386
	Std. Dev.	0.651	0.653	0.581	0.495
DG P (MW)	Mean	–	–	0.282	0.616
	Max	–	–	1.348	1.362
	Std. Dev.	–	–	0.389	0.359
DG Q (MVar)	Mean	–	–	0.119	0.260
	Max	–	–	1.962	1.982
	Std. Dev.	–	–	0.301	0.406

Table S3: Statistical Information of the IEEE 69-bus Network Input Datasets

Parameter	Indices	ISM	RF	RF+DG	NST+DG
V (p.u.)	Min	0.901	0.881	0.821	0.829
	Mean	0.975	0.974	0.975	0.985
	Max	1.000	1.000	1.035	1.046
	Std. Dev.	0.025	0.027	0.028	0.029
Load P (MW)	Mean	18.470	19.956	24.913	30.320
	Max	37.960	37.960	79.211	80.003
	Std. Dev.	4.264	5.199	11.482	13.916
Load Q (MVar)	Mean	53.243	55.362	54.230	53.594
	Max	65.772	73.760	95.346	96.299
	Std. Dev.	7.264	8.195	11.211	13.768
DG P (MW)	Mean	–	–	4.218	8.649
	Max	–	–	23.316	23.549
	Std. Dev.	–	–	5.697	5.384
DG Q (MVar)	Mean	–	–	1.107	2.271
	Max	–	–	57.877	58.456
	Std. Dev.	–	–	5.878	8.341

2 Tables from results section

The tables for results section are reproduced here in tables S4 and S5 in case exact numerical values are required by the readers.

References

- [1] F. Jabari, F. Sohrabi, P. Pourghasem, and B. Mohammadi-Ivatloo, “Backward-Forward Sweep Based Power Flow Algorithm in Distribution Systems,” *International Journal of Electrical Power & Energy Systems*, vol. 123, Art. no. 106325, 2020.
- [2] A. A. Şeker, T. Gözel, and M. H. Hocaoglu, “BIBC Matrix Modification for Network Topology Changes: Reconfiguration Problem Implementation,” *International Journal of Electrical Power & Energy Systems*, vol. 131, Art. no. 107054, 2021.

Table S4: Percentage Accuracy (PAQ) Comparison

Algorithm (PAQ%)	Case	Percentage Accuracy (PAQ%)				Overall PAQ
		<i>T</i>	<i>R</i>	<i>X</i>	θ	
30B <i>VMDTI</i> (94.5)	<i>ISM</i>	100	99.99	99.89	99.87	99.9
	<i>RF</i>	91.1	96.27	93.16	99.13	94.9
	<i>RF+DG</i>	91.1	96.27	90.93	100	94.6
	<i>NST+DG</i>	94.5	93.54	75.32	90.3	88.4
30B <i>MOTE</i> (96.8)	<i>ISM</i>	100	99.99	99.94	99.94	100.0
	<i>RF</i>	94.8	98.63	97.17	99.71	97.6
	<i>RF+DG</i>	94.8	98.56	96.33	100	97.4
	<i>NST+DG</i>	95.2	95.62	84.46	94.3	92.4
33B <i>VMDTI</i> (92.1)	<i>ISM</i>	100	99.27	97.9	88.58	96.4
	<i>RF</i>	94.9	88.14	86.57	93.18	90.7
	<i>RF+DG</i>	97	88.73	87.1	98.08	92.7
	<i>NST+DG</i>	96.6	91.04	88.6	78.11	88.6
33B <i>MOTE</i> (94.3)	<i>ISM</i>	100	99.52	98.59	92.08	97.5
	<i>RF</i>	95.4	92.8	91.7	92.73	93.2
	<i>RF+DG</i>	97	92.19	91.01	98.71	94.7
	<i>NST+DG</i>	96.6	93.84	92.1	84.25	91.7
69B <i>VMDTI</i> (92.1)	<i>ISM</i>	100	99.99	99.98	99.88	100.0
	<i>RF</i>	92.2	88.89	88.89	87.98	89.5
	<i>RF+DG</i>	91.6	88.89	88.89	87.98	89.3
	<i>NST+DG</i>	92	88.89	88.89	88.71	89.6
69B <i>MOTE</i> (95.2)	<i>ISM</i>	100	100	99.99	99.92	100.0
	<i>RF</i>	98.8	94.9	93.43	99.75	96.7
	<i>RF+DG</i>	91.6	92.31	92.31	91.65	92.0
	<i>NST+DG</i>	92	92.31	92.31	92.18	92.2
Overall PAQ of the <i>VMDTI</i>						92.89
Overall PAQ of the <i>MOTE</i>						95.45

Table S5: Mean Absolute Error (MAE) Comparison

Algorithm (MAE $\times 10^{-3}$)	Case	Mean Absolute Error ($\times 10^{-3}$)				Overall MAE
		<i>T</i>	<i>R</i>	<i>X</i>	θ	
30B <i>VMDTI</i> (30.3)	<i>ISM</i>	0.0	0.0	0.0	1.3	0.3
	<i>RF</i>	88.5	5.7	2.2	8.9	26.3
	<i>RF+DG</i>	88.5	5.7	3.0	0.0	24.3
	<i>NST+DG</i>	55.2	10.2	9.9	205.9	70.3
30B <i>MOTE</i> (18.1)	<i>ISM</i>	0.0	0.0	0.0	0.7	0.2
	<i>RF</i>	51.7	2.1	0.9	2.9	14.4
	<i>RF+DG</i>	51.7	2.2	1.2	0.0	13.8
	<i>NST+DG</i>	48.3	6.8	5.6	115.9	44.1
33B <i>VMDTI</i> (43.1)	<i>ISM</i>	0.0	3.1	7.8	71.3	20.5
	<i>RF</i>	51.1	57.0	56.4	40.4	51.2
	<i>RF+DG</i>	30.3	53.8	53.8	10.8	37.2
	<i>NST+DG</i>	34.1	41.7	46.7	131.4	63.5
33B <i>MOTE</i> (31.3)	<i>ISM</i>	0.0	2.1	5.2	47.5	13.7
	<i>RF</i>	46.4	32.9	32.9	43.3	38.9
	<i>RF+DG</i>	30.3	35.9	35.9	7.2	27.3
	<i>NST+DG</i>	34.1	27.8	31.1	87.6	45.2
69B <i>VMDTI</i> (32.9)	<i>ISM</i>	0.0	0.0	0.0	0.4	0.1
	<i>RF</i>	78.4	2.7	1.3	41.4	31.0
	<i>RF+DG</i>	84.2	2.7	1.3	41.4	32.4
	<i>NST+DG</i>	80.1	2.7	1.3	188.4	68.1
69B <i>MOTE</i> (21.1)	<i>ISM</i>	0.0	0.0	0.0	0.2	0.1
	<i>RF</i>	11.7	1.2	0.7	0.8	3.6
	<i>RF+DG</i>	84.2	1.8	0.9	27.6	28.6
	<i>NST+DG</i>	80.1	1.8	0.9	125.6	52.1
Overall MAE of the <i>VMDTI</i>						35.44
Overall MAE of the <i>MOTE</i>						23.49